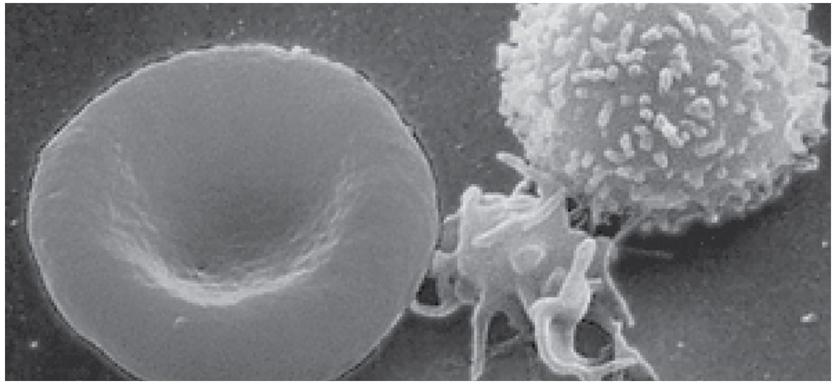
CHAPTER 18

The Cardiovascular System: Blood



Plasma, blood cells, clotting, and blood types.

Anatomy & Physiology · Broward-Miami Health Institute

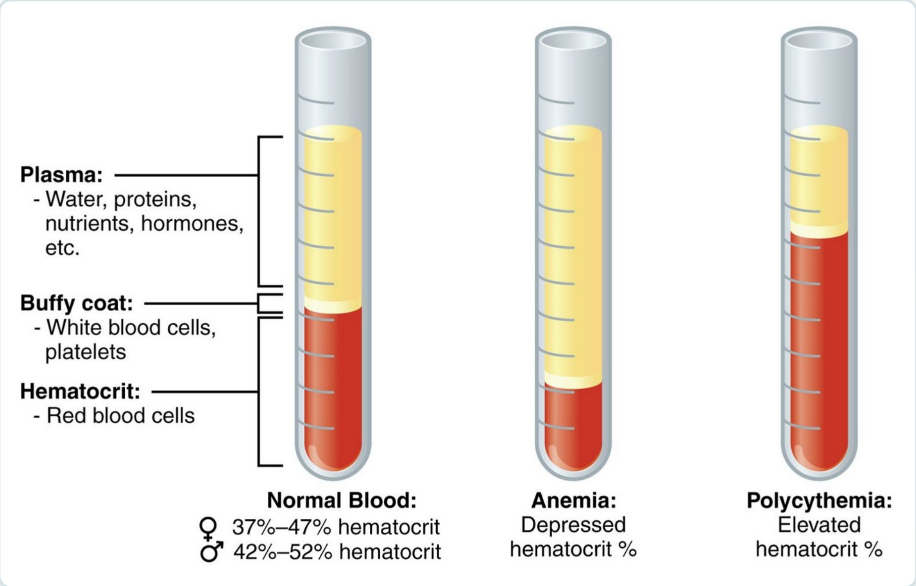
FACULTY EDITION

CHAPTER 18

Learning Objectives

After studying this chapter, students will be able to:

1. Identify the primary functions of blood, its fluid and cellular components, and its physical characteristics
2. Identify the most important proteins and other solutes present in blood plasma
3. Describe the formation of the formed element components of blood
4. Discuss the structure and function of red blood cells and hemoglobin
5. Classify and characterize white blood cells
6. Describe the structure of platelets and explain the process of hemostasis



Blood: plasma and formed elements.

18.1

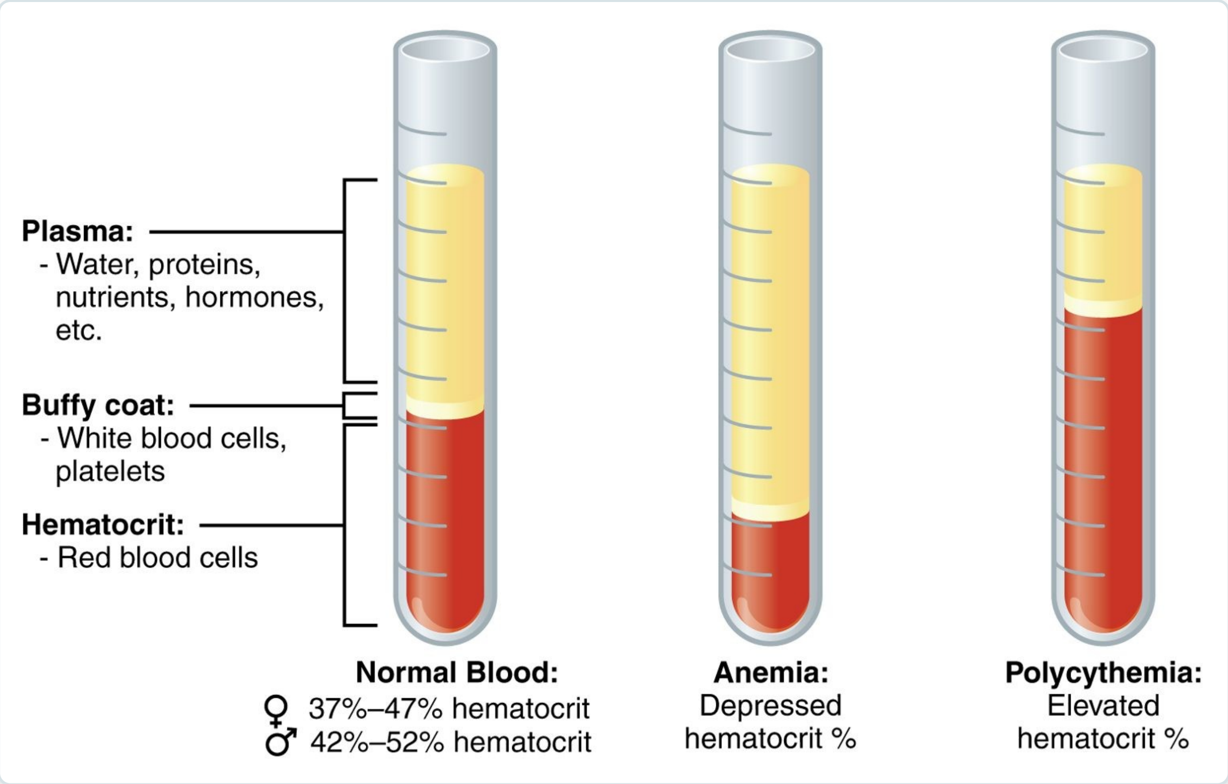
SECTION

An Overview of Blood

What blood is made of and what it does.

SECTION 18.1

Composition of Blood



Blood: plasma and formed elements

- Plasma is the liquid portion (~55%)
- Formed elements are cells (~45%)
- Buffy coat = white cells and platelets
- Hematocrit = percent red cells

SECTION

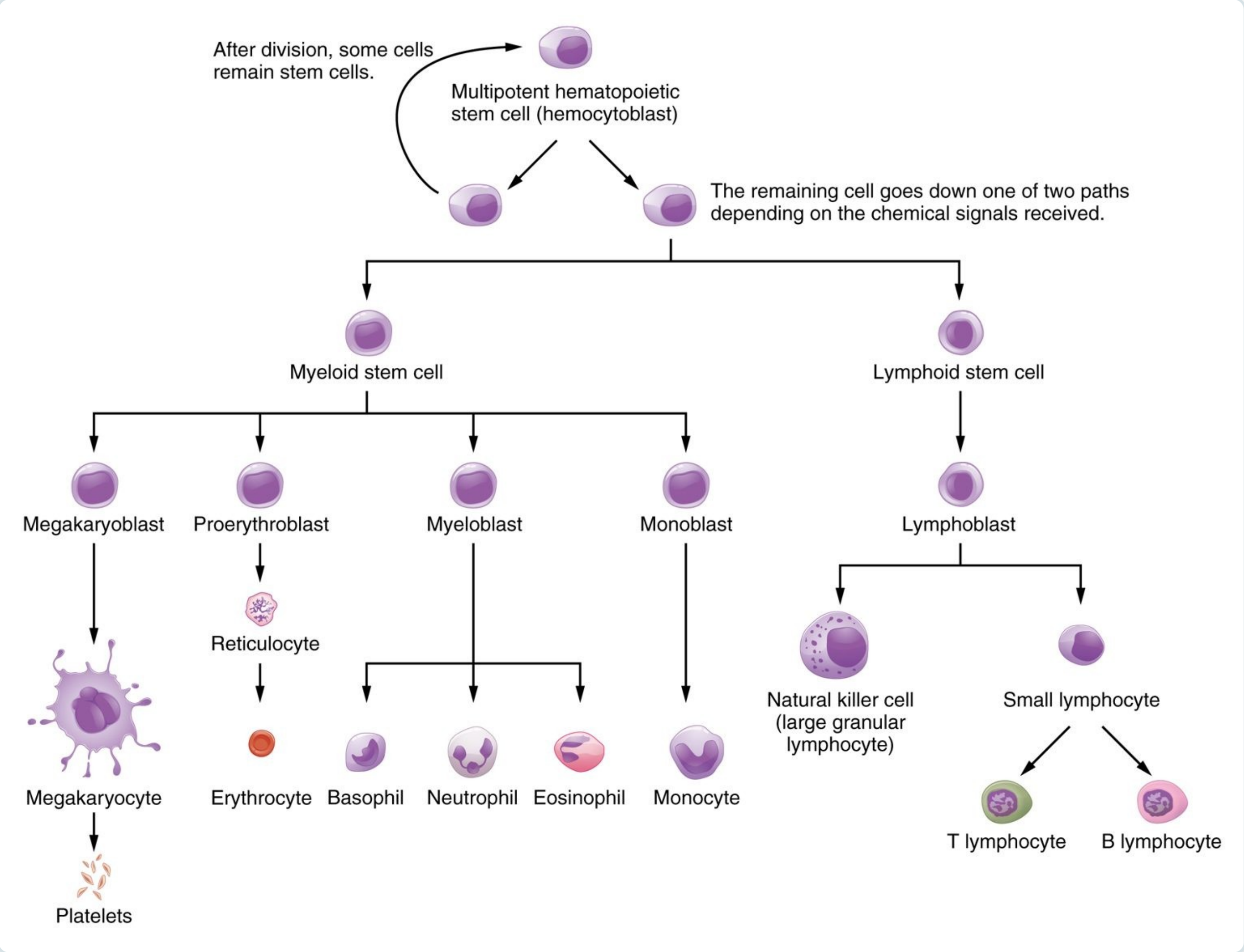
18.2

Production of the Formed Elements

How blood cells are produced in red bone marrow.

Hematopoiesis

- All blood cells form in red marrow
- Start from one stem cell
- Myeloid line → most cell types
- Lymphoid line → lymphocytes



Hematopoiesis: the formation of blood cells

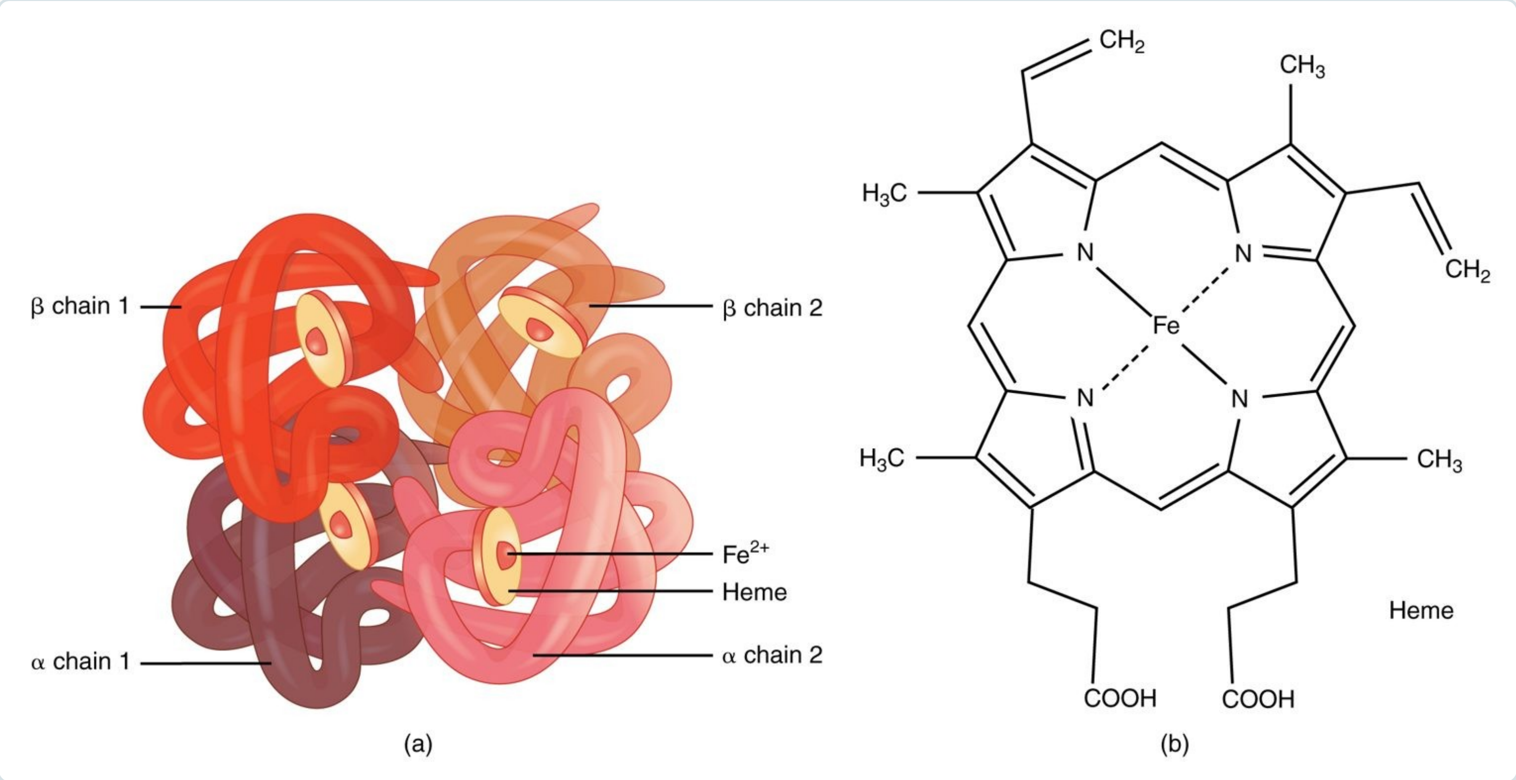
SECTION

18.3

Erythrocytes

Red blood cells and oxygen transport.

Hemoglobin

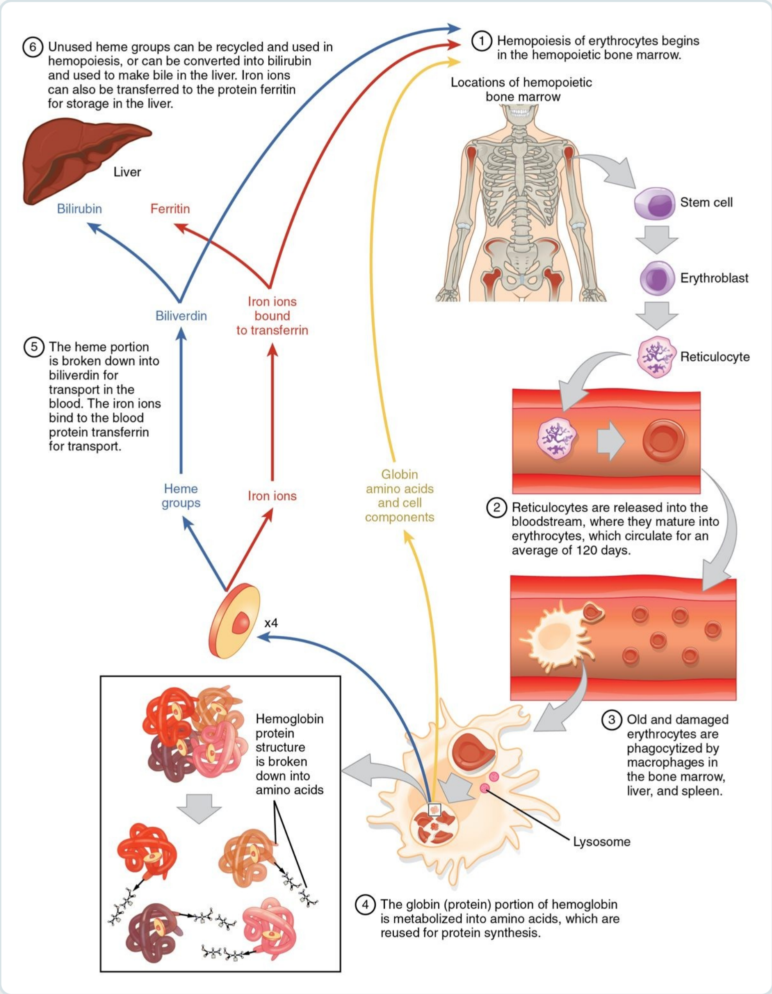


The structure of hemoglobin

- The oxygen-carrying protein in red cells
- Four globin chains
- Each holds an iron-rich heme group
- Iron binds the oxygen

SECTION 18.3

Red Blood Cell Life Cycle



The life cycle of a red blood cell

- Red cells live about 120 days
- Old cells broken down in spleen and liver
- Iron is recycled
- Heme becomes bilirubin

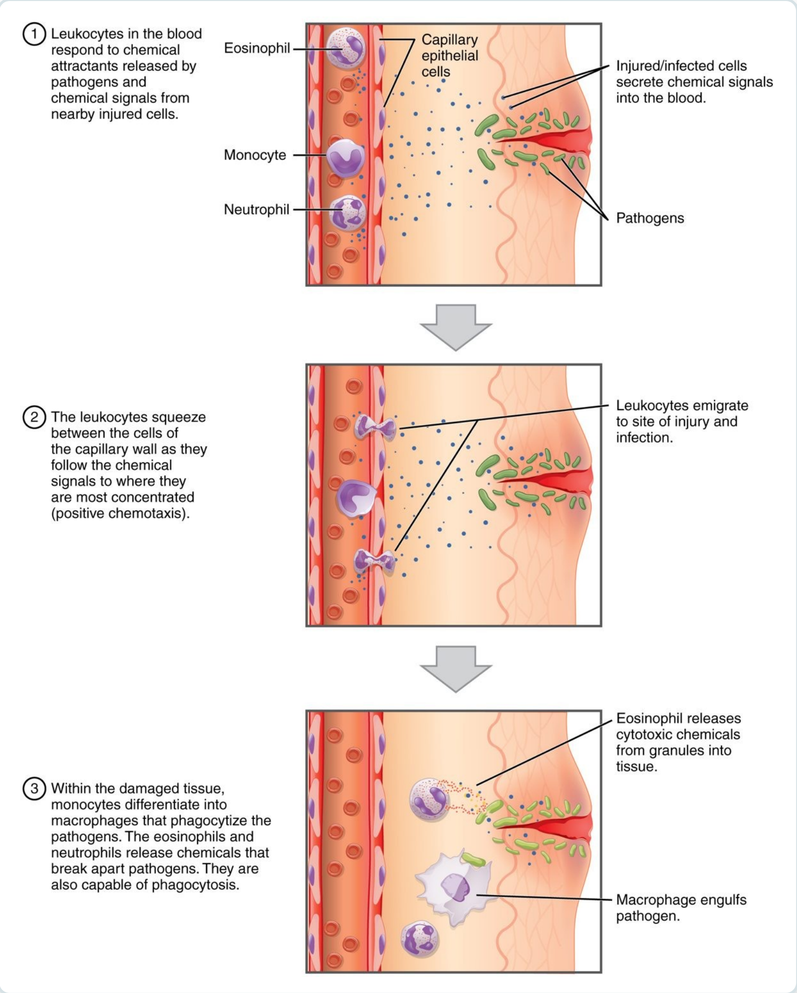
18.4

SECTION

Leukocytes and Platelets

White blood cells and platelets — defense and clotting.

How Leukocytes Fight Infection



How leukocytes fight infection

- White cells defend against infection
- Leave vessels by diapedesis
- Move toward chemical signals
- Engulf pathogens (phagocytosis)

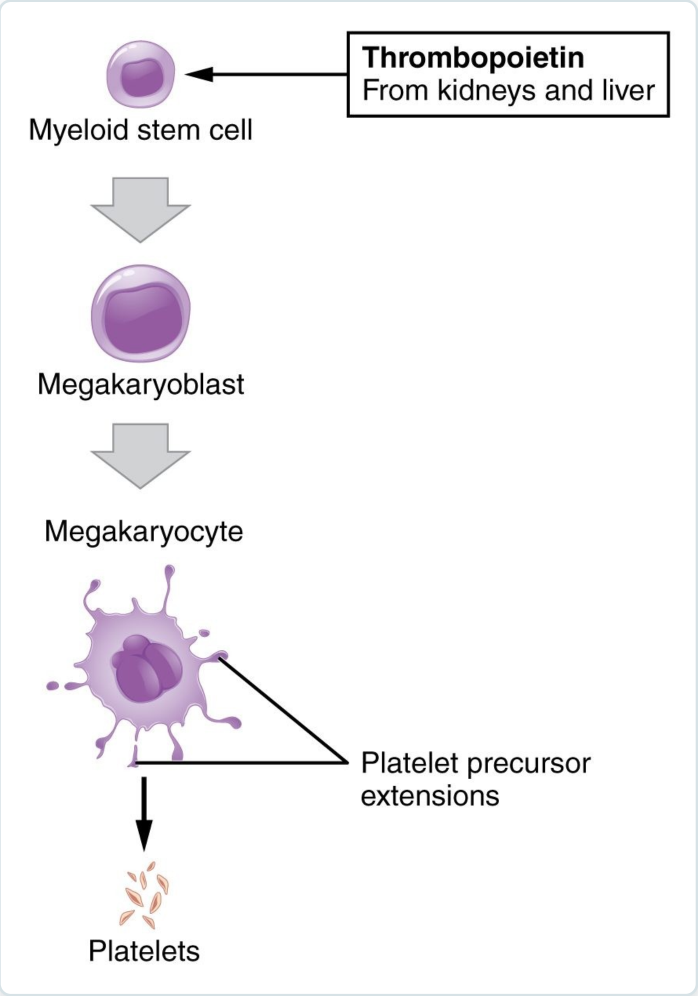
Types of White Blood Cells

The granular leukocytes

- Five types of leukocytes
- Granulocytes: neutrophil, eosinophil, basophil
- Agranulocytes: lymphocyte, monocyte
- Each has a specific role

SECTION 18.4

Platelet Formation



The formation of platelets

- Platelets are cell fragments
- Shed from megakaryocytes
- Driven by thrombopoietin
- Essential for clotting

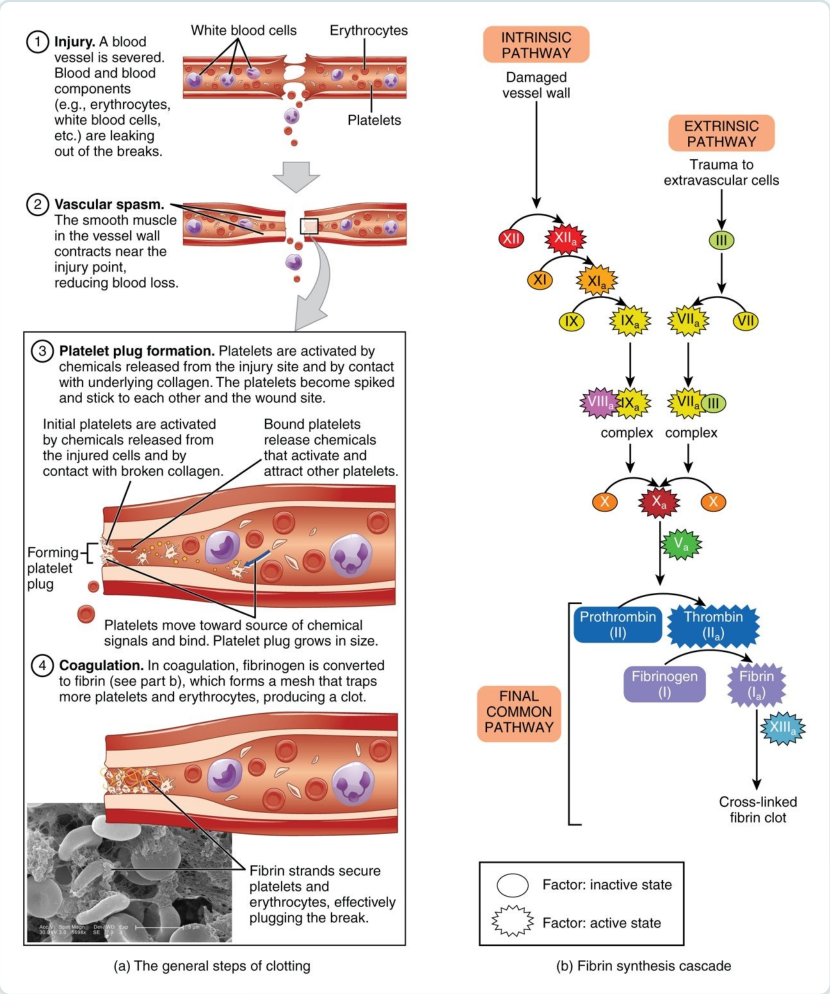
18.5

SECTION

Hemostasis

Hemostasis — how the body stops bleeding.

Hemostasis



The steps of hemostasis

- Stops bleeding in three steps
- Vascular spasm narrows the vessel
- A platelet plug forms
- Coagulation makes a fibrin clot

18.6

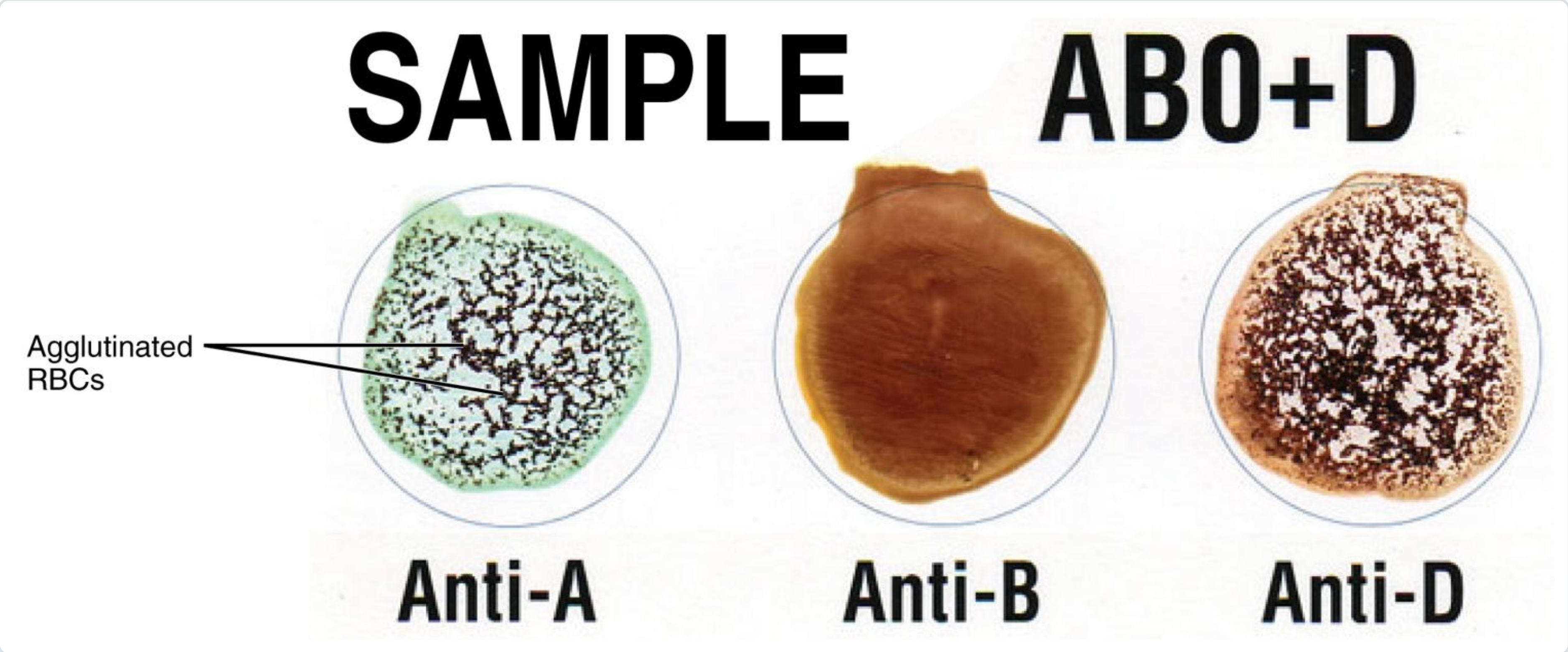
SECTION

Blood Typing

Blood typing and transfusion compatibility.

SECTION 18.6

Blood Typing



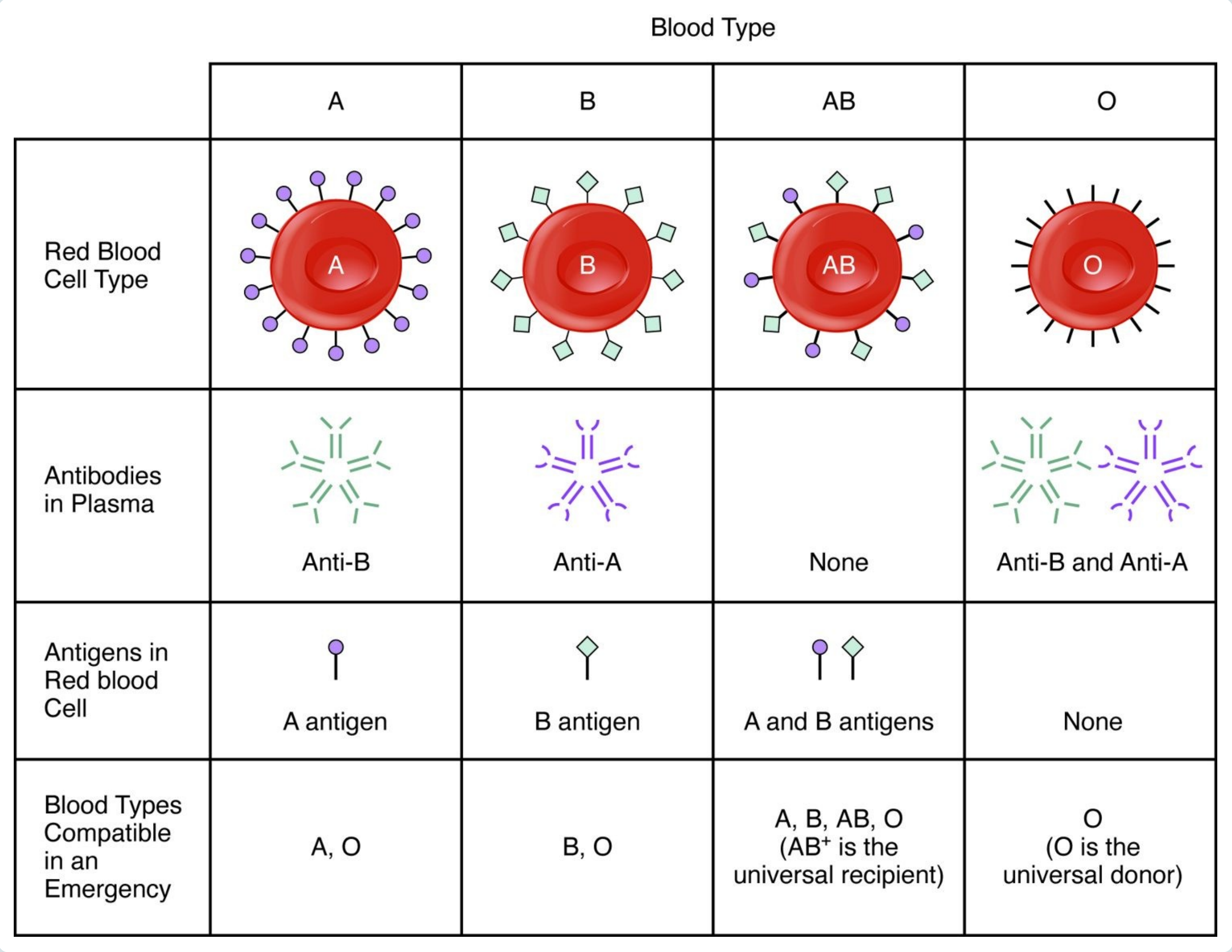
Blood typing by agglutination

- Type set by red-cell surface antigens
- ABO system: A, B, AB, O
- Rh system: positive or negative
- Tested by agglutination

SECTION 18.6

Transfusion Compatibility

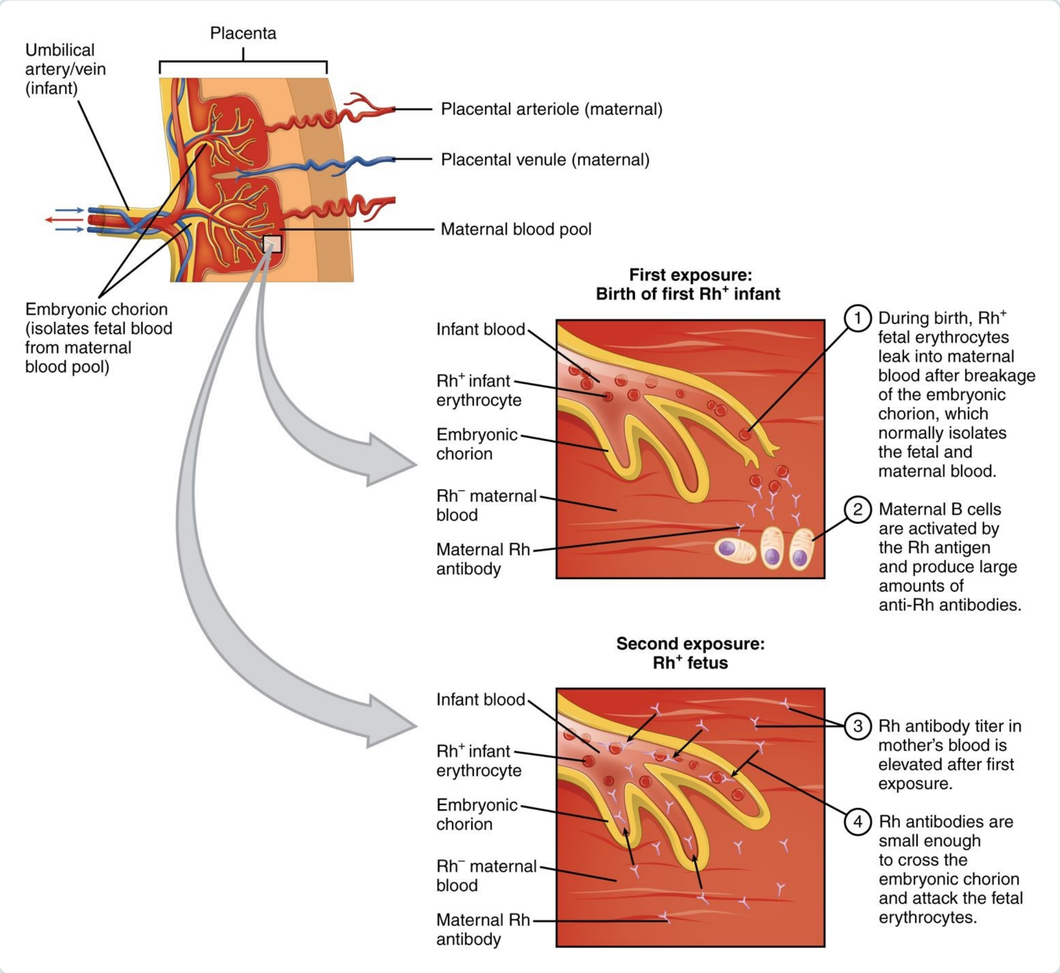
- Plasma holds antibodies to missing antigens
- The wrong type causes agglutination
- Type O-negative is universal donor
- Type AB-positive is universal recipient



ABO blood type compatibility

Rh Incompatibility

- Rh-negative mother, Rh-positive baby
- First exposure sensitizes the mother
- Antibodies threaten a later baby
- Prevented with Rh immune globulin



Rh incompatibility in pregnancy

Key Takeaways

- Blood is plasma plus formed elements — red cells, white cells, and platelets.
- All blood cells form in red bone marrow from a single stem cell.
- Red cells carry oxygen on hemoglobin and are recycled after about 120 days.
- White cells defend against infection; platelets and clotting factors stop bleeding.
- Blood type (ABO and Rh) must be matched before transfusion.

CLINICAL CONNECTIONS

Why Blood Matters in Practical Nursing

Anemia

Too few red cells or too little hemoglobin reduces oxygen delivery.

Complete Blood Count

The CBC is one of the most common diagnostic tests.

Transfusion Safety

Matching ABO and Rh type prevents dangerous reactions.

Clotting Disorders

Too little clotting bleeds; too much causes thrombosis.

Anticoagulants

Drugs like heparin and warfarin act on the clotting cascade.

Infection

A rising white cell count is an early sign of infection.